Quantum data parallelism in quantum neural networks

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Quantum neural networks hold promise for achieving lower generalization error bounds and enhanced computational efficiency in processing certain datasets. However, the integration of quantum superposition in data parallelism—a key aspect in the applications of classical neural network—has been largely unexplored in the context of quantum neural networks. Here, we demonstrate the effective application of quantum parallelism, via quantum superposition and entanglement, to achieve data parallelism in generic quantum neural network models. We address two classes of encoding schemes for the training data: as orthogonal and nonorthogonal quantum states, respectively. With orthogonal encoding, we rigorously prove that parallel and individual training are equivalent in terms of the loss function, a result directly derived from quantum entanglement; with nonorthogonal encoding, we show that parallel computing remains viable, and quantum interference among data becomes notable. We perform a comparative numerical analysis in these different scenarios, exploring various quantum neural networks with or without superposed data input, highlighting the validity of quantum data parallelism.

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I. INTRODUCTION

Quantum parallelism, the capability to execute numerous operations concurrently by exploiting quantum physics, stands as a fundamental advantage of quantum computing over classical computing. This principle is vital in many prominent quantum algorithms. A prime example is Shor's algorithm [1], which can be used to defeat the Rivest-Shamir-Adleman (RSA) encryption. Shor's algorithm reduces the factorization problem to finding the period of a function, utilizing quantum parallelism to simultaneously evaluate all function values in a single step [2,3]. Other well-known quantum algorithms, such as the Deutsch-Jozsa algorithm [4], the quantum Fourier transform [5], and Grover's search algorithm [6], also harness this unique capability of quantum computing. By enabling a quantum computer to explore multiple classical paths all at once, these algorithms provide prototypes for the distinct advantages of quantum parallelism.

The development of quantum neural networks (QNNs) has emerged as a dynamic area of interest, offering new perspectives and possibilities in machine learning [7–9]. In the current context, we adopt a broad definition of QNNs, recognizing them as a subset of quantum circuits containing parametrized gates. The dominant training method for QNNs involves passing a data point through a parametrized quantum circuit, contributing to the loss function via measurement, with the loss typically computed after a batch or the entire dataset is processed [10–12]. This approach, however, requires repeatedly invoking quantum processing of each data point, a method that has overlooked the potential of data parallelism with quantum principles.

In classical machine learning, data parallelism refers to the technique where multiple processors, such as GPUs, simultaneously train the same model on different subsets of data. The exercise of data parallelism is crucial and significantly accelerates the training process, particularly in the context of large-scale models [13–16]. This naturally leads to a key question: can data parallelism be effectively implemented with QNNs? In classical parallelism, data are processed in parallel but still separately in the physical sense. In contrast, quantum parallelism allows for the simultaneous processing of multiple data within a single physical system, which is inherently more integrated and more efficient.

Indeed, quantum data parallelism has been touched upon in some studies [17,18] with training data points represented by orthogonal quantum states. However, these works neither focus on this aspect nor provide a systematic study by comparing parallel training with traditional individual training methods. Additionally, specific QNN models designed for parallel computation have been proposed [19–21], albeit with limitations. For instance, in binary classification, the models by Adhikary *et al.* [19] and Zhang *et al.* [20] enable the simultaneous training of only two data points from different labels, while Dangwal *et al.* [21] assume a balanced distribution of two labels within the training set and introduce additional control qubits to manage the number of parallel data points. Notably, to the best of our knowledge, the schemes in which training data points are encoded as

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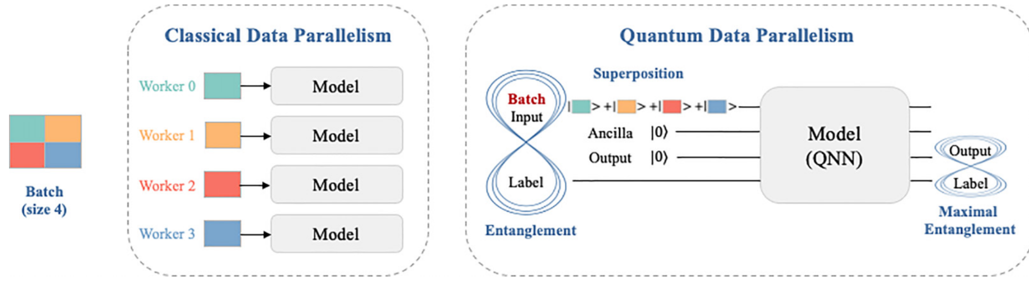

FIG. 1. Conceptual diagram of quantum data parallelism in QNNs. In classical data parallelism, workers handle data in parallel, either within a single machine (e.g., multicore processors) or across multiple machines (e.g., clusters), but each worker processes data independently. In contrast, quantum parallelism offers a more integrated and efficient framework, which can lead to interactions among data elements. We demonstrate how encoding a batch of data in a quantum superposition state facilitates the parallel training of QNN models via data-label entanglement. Here, the training of a QNN is achieved by maximizing the possible entanglement between the output and label qubits.

nonorthogonal quantum states remain unexplored in the context of QNNs.

To bridge the gap, we unveil how the key quantum principles, namely superposition and entanglement, enable data parallelism in generic QNNs (see Fig. 1). We start with exactly solvable examples based on the XOR problem, examining the differences between individual and parallel trainings with orthogonal or nonorthogonal input quantum states. Through analytical proofs and numerical simulations that solve the XOR classification task, we illustrate the decision boundary of the quantum circuit in handling this toy problem, providing a general intuition about parallel data processing in QNNs. Extending along these lines, we propose a generic QNN framework that includes most currently used QNN models, within which we conduct a detailed theoretical analysis on how data parallelism works by virtue of quantum superposition. We rigorously prove that, in the orthogonal case, parallel and traditional training methods are equivalent in terms of the loss function, which lends an immediate advantage to the parallel approach in data processing. In the nonorthogonal case, we show that parallel training remains viable with quantum interference between data becoming substantial. Our results, validated using real-world datasets such as Iris and MNIST, establish the feasibility of quantum data parallelism in QNNs.

II. EXACTLY SOLVABLE EXAMPLE

To illustrate quantum data parallelism in QNNs, we first consider an exactly solvable problem—the XOR problem. This binary function takes two binary inputs and outputs true only when these inputs differ, as depicted in Fig. 2(a). The XOR problem is historically significant as it highlighted the limitations of early neural networks [22], leading to significant advancements in the field. Addressing this problem typically necessitates implementing multilayer neural networks to introduce the required nonlinearity [23]. In the context of quantum circuits, classical data are encoded into quantum states, naturally mapping them to a higher-dimensional Hilbert space in a nonlinear way, which is unattainable by classical linear models. Its simple yet nontrivial nature, along with our chosen quantum circuit, makes the XOR problem an ideal example, allowing for direct comparison with classical neurons while simultaneously highlighting the unique advantage of quantum parallelism.

A. Orthogonal encoding

In the orthogonal case, we utilize basis encoding $\{|x_1^i x_2^i\rangle|y^i\rangle\}_{i=1}^4 = \{|00\rangle|0\rangle, |01\rangle|1\rangle, |10\rangle|1\rangle, |11\rangle|0\rangle\}$ to represent the four training data points, and we consider a quantum circuit inspired by the concept of a quantum neuron [18,24,25], depicted in Fig. 2(b). This circuit's conceptual similarity to a classical neuron facilitates comparative analysis. For each training data, the measurement $M = -Z_{\hat{y}}Z_y = -\mathrm{diag}(1, -1, -1, 1)$ on the output qubit and the label qubit gives a result between -1 and 1. When $\langle M\rangle < 0$, it indicates that the data point has been correctly classified.

1. Individual input

The goal of the training is to minimize the loss function over the training dataset,

$$L_I^O = \frac{1}{4}\sum_i^4 \langle \hat{y}^i y^i|M|\hat{y}^i y^i\rangle, \tag{1}$$

where $|\hat{y}^i\rangle = \cos\frac{1}{2}(w_1x_1^i + w_2x_2^i + b)|0\rangle + \sin\frac{1}{2}(w_1x_1^i + w_2x_2^i + b)|1\rangle$ is the final state of the output qubit [see Fig. 2(b)]. By substituting the set of four data points $\{|x_1^i x_2^i\rangle|y^i\rangle\}_{i=1}^4$ into Eq. (1), we obtain

$$\begin{aligned} L_I^O = &-\tfrac{1}{4}[\cos b - \cos(w_1 + b) \\ &- \cos(w_2 + b) + \cos(w_1 + w_2 + b)]. \end{aligned} \tag{2}$$

When $w_1 = w_2 = \pi$ and $b = 0$, the loss L_I^O reaches its minimum value of -1, achieving a perfect classification of all four data points.

2. Batch input

For a parallel training with multiple data points, we employ an equal-weight superposition state $\frac{1}{2}\sum_i^4 |x_1^i x_2^i\rangle|y^i\rangle$ as the input. Subsequently, we use individual data points $\{|x_1^i x_2^i\rangle|y^i\rangle\}_{i=1}^4$ to test the trained circuit. The goal of the training is to minimize

$$L_B^O = \langle\psi|\mathbb{1}_x M|\psi\rangle, \tag{3}$$

where $|\psi\rangle = \frac{1}{2}\sum_i^4 |x_1^i x_2^i\rangle|\hat{y}^i\rangle|y^i\rangle$. Unlike the individual input case, $|\psi\rangle$ is a highly entangled state. By evaluating the above expression as a function of w and b, we find that $L_B^O = L_I^O$.

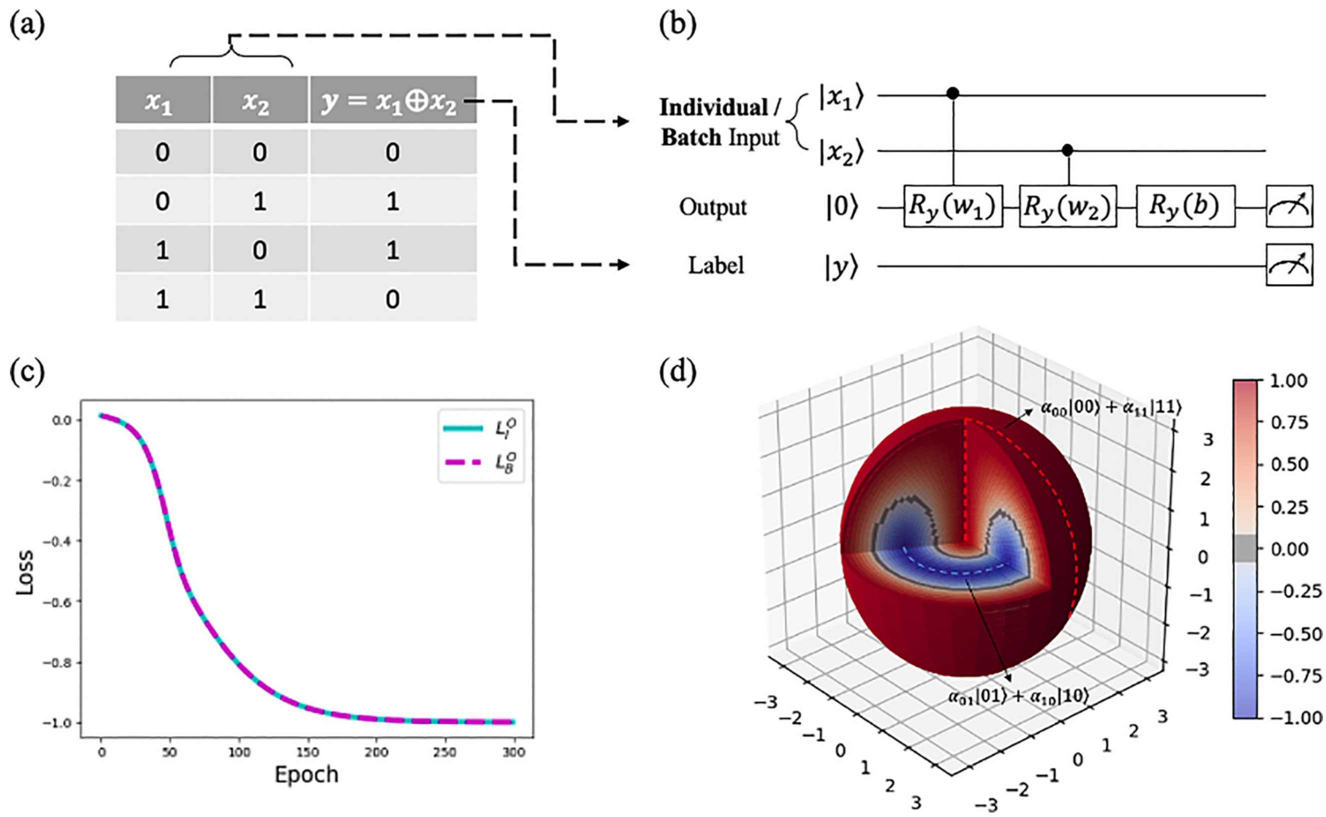

FIG. 2. Solving the XOR problem via a minimal quantum circuit with orthogonal encoding. (a) XOR Truth table. (b) Quantum circuit employed in this case. Here we use the convention $R_p(w) = \exp(-iPw/2)$, where $p \in \{x, y, z\}$ labels Pauli operators $P \in \{X, Y, Z\}$, respectively. (c) Evolution of the individual training loss L_I^O [see Eq. (1)] and the parallel training loss L_B^O [see Eq. (3)] over epochs. The two curves fall exactly on top of each other, which agrees with the theoretical conclusion that $L_I^O = L_B^O$ (see Theorem 1). (d) Visualization of the decision boundary derived from the trained quantum circuit. Here we map two-qubit states $|x_1x_2\rangle = \sum_{i,j=0}^{1} \alpha_{ij}|ij\rangle$, where $\alpha_{ij} \in \mathbb{R}$ and $\sum_{i,j=0}^{1} \alpha_{ij}^2 = 1$, to a 3-sphere represented by a solid ball with its surface identified as a single point (the image of $|11\rangle$). The color of each point in the 3-sphere is determined by the circuit's prediction when the represented state is the input. To better illustrate the decision boundary, a segment of the solid ball is excised, and two dashed lines are added to indicate the subspaces spanned by the two classes of states, with the gray region representing the decision boundary.

Consequently, L_B^O also reaches its minimum value of -1 when $w_1 = w_2 = \pi$ and $b = 0$.

We simulate the quantum circuit in Fig. 2(b) with both the individual and the batch input by using PennyLane [26]. In both scenarios, we employ gradient descent for optimization, and we use the same initialization parameters. The numerical results are presented in Fig. 2(c). In this figure, the curves representing the decline of individual training loss L_I^O [see Eq. (1)] and parallel training loss L_B^O with epochs completely overlap, consistent with the theoretical equivalence of $L_B^O = L_I^O$. Notably, both curves converge to a minimum value of -1 ($w_1 = w_2 = \pi$ and $b = 0$), indicating that the trained circuit can classify each individual data point perfectly. This observation suggests that, even when trained using a superposition state comprising multiple data points, the trained circuit retains the capability to accurately classify individual data $\{|x_1^i x_2^i\rangle|y^i\rangle\}_{i=1}^4$.

To see how the quantum circuit makes its decisions, we visualize the decision boundary of the trained quantum circuit. First, we traverse all two-qubit states $|x_1x_2\rangle = \sum_{i,j=0}^{1} \alpha_{ij}|ij\rangle$, where $\alpha_{ij} \in \mathbb{R}$, using them as inputs to the trained quantum circuit. We then measure the output qubit $|\hat{y}\rangle$ and use the expected value of the measurement as the prediction result. The measurement $Z_{\hat{y}} = \mathrm{diag}(1, -1)$ on the output qubit gives the result $\langle\hat{y}|Z_{\hat{y}}|\hat{y}\rangle \in [-1, 1]$, where -1 and 1 correspond to $|\hat{y}\rangle = |1\rangle$ and $|\hat{y}\rangle = |0\rangle$, respectively. The set of points $\{(\alpha_{00}, \alpha_{01}, \alpha_{10}, \alpha_{11}) \in \mathbb{R}^4 : \alpha_{00}^2 + \alpha_{01}^2 + \alpha_{10}^2 + \alpha_{11}^2 = 1\}$ form a unit 3-sphere S^3, which can be parametrized using three coordinates as follows:

$$\alpha_{00} = \sin\zeta_1 \cos\zeta_2, \quad \alpha_{01} = \sin\zeta_1 \sin\zeta_2 \cos\zeta_3,$$
$$\alpha_{10} = \sin\zeta_1 \sin\zeta_2 \sin\zeta_3, \quad \alpha_{11} = \cos\zeta_1, \tag{4}$$

where $\zeta_1 \in [0, \pi]$, $\zeta_2 \in [0, \pi]$, and $\zeta_3 \in [0, 2\pi)$. By denoting ζ_1 as the radius, ζ_2 as the polar angle, ζ_3 as the azimuthal angle, and representing the measurement value through color, we visualize this configuration as a solid ball, as depicted in Fig. 2(d). In this representation, the surface of the solid sphere is identified as a single point. In the figure, the effectiveness of the trained quantum circuit in addressing the XOR problem is clearly demonstrated. For instance, the color value on both the axis and the surface of the solid sphere, highlighted by the red circle, is approximately 1, indicating that when the input is $|x_1x_2\rangle = \alpha_{00}|00\rangle + \alpha_{11}|11\rangle$, the trained quantum circuit

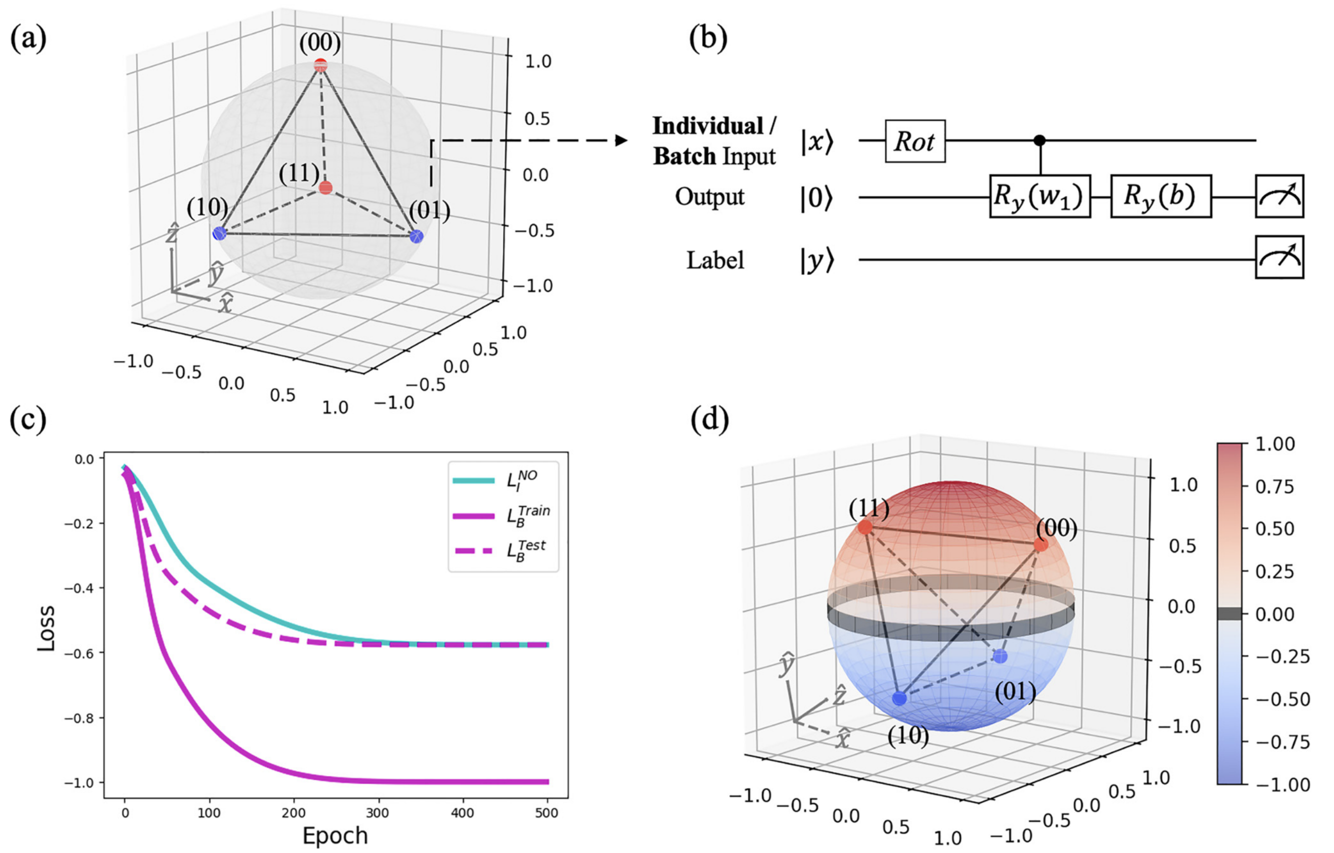

FIG. 3. Solving the XOR problem via a minimal quantum circuit with nonorthogonal encoding. (a) Representation of four data points by the four vertices of a regular tetrahedron on the Bloch sphere. (b) Quantum circuit schematic, where Rot $= R_z(\xi_1)R_y(\xi_2)R_z(\xi_3)$ is an arbitrary single-qubit rotation. (c) Evolution of the individual training loss L_I^{NO} [see Eq. (5)], the parallel training loss L_B^{Train} [see Eq. (7)], and the individual testing loss L_B^{Test} [see Eq. (9)] over epochs. (d) Visualization of the decision boundary derived from the trained quantum circuit. The color of each point on the Bloch sphere is determined by the circuit's prediction when the represented state is the input. The gray band around the equator represents the decision boundary between the two classes. Note that the Rot gate maintains the tetrahedron configuration while mapping the four data points to different positions on the Bloch sphere.

produces an output of $|0\rangle$, consistent with the label. In contrast, a single classical neuron using common activation functions like ReLU cannot solve the XOR problem, which typically requires multilayer neural networks. Our circuit projects data into a higher dimension and a different topology from the representation space for a classical neuron. Furthermore, the ability to process multiple data points in parallel is also a unique characteristic of quantum circuits.

B. Nonorthogonal encoding

In the nonorthogonal case, we adopt a quantum circuit [see Fig. 3(b)] similar to the one used in the orthogonal case. While we continue to perform the $M = -Z_{\hat{y}}Z_y$ measurement on the output and label qubit, we adopt a distinct encoding method.

1. Individual input

When using the basis encoding, the four data points of the XOR problem are orthogonal, implying they are equidistant from each other. We preserve this property in the nonorthogonal case by encoding the four data points as the vertices of a regular tetrahedron within the Bloch sphere, $\{|x^i\rangle\}_{i=1}^4$. Subsequently, both the data and its label $\{|x^i\rangle|y^i\rangle\}_{i=1}^4$ are used as training data for the quantum circuit. We adopt the following loss function:

$$L_I^{\mathrm{NO}} = \frac{1}{4}\sum_i^4 \langle \hat{y}^i y^i|M|\hat{y}^i y^i\rangle, \tag{5}$$

where $|\hat{y}\rangle$ is the final state of output qubits. To evaluate the minimum value of L_I^{NO}, we parametrize the quantum state of $|x^i\rangle$ post the Rot gate to be $\cos\frac{\theta_i}{2}|0\rangle + e^{i\varphi_i}\sin\frac{\theta_i}{2}|1\rangle$ without loss of generality. Incorporating this into Eq. (5) yields

$$\begin{aligned} L_I^{\mathrm{NO}} = &-\tfrac{1}{8}\{(\cos\theta_1 - \cos\theta_2 - \cos\theta_3 + \cos\theta_4) \\ &\times [\cos b - \cos(w_1 + b)]\}. \end{aligned} \tag{6}$$

The constraint for minimizing this expression is that $\{\theta_i\}_{i=1}^4$ correspond to the polar angle of the four vertices of the regular tetrahedron. When $\cos\theta_1 = -\cos\theta_2 = -\cos\theta_3 = \cos\theta_4 = \pm\frac{1}{\sqrt{3}}$, and with $b = 0$, $w_1 = \pi$, or $b = w_1 = \pi$, the expression Eq. (6) reaches its minimum value $-\frac{1}{\sqrt{3}}$ (see Appendix A).

2. Batch input

Utilizing the tetrahedral representation of data points $\{|x^i\rangle\}_{i=1}^4$, we can also process multiple data entries in parallel by constructing a superposed state $[(|x^1\rangle + |x^4\rangle)|00\rangle + (|x^2\rangle + |x^3\rangle)|01\rangle]/N$, where $N^2 = 4 + 2\,\mathrm{Re}(\langle x^1|x^4\rangle) + 2\,\mathrm{Re}(\langle x^2|x^3\rangle)$ is the normalization factor. Note that, while the position of each state on the Bloch sphere is predetermined, the relative phase between these states is variable. Consistent with the tetrahedron's constraints, the inner products are determined as $\langle x^1|x^4\rangle = \frac{1}{\sqrt{3}}e^{i\gamma_{14}}$ and $\langle x^2|x^3\rangle = \frac{1}{\sqrt{3}}e^{i\gamma_{23}}$. Here, the phases γ_{14} and γ_{23} are adjustable and have a direct impact on both N and the loss function. We employ this superposition state as the training input and adopt the following loss function:

$$L_B^{\text{Train}} = \langle\psi'|\mathbb{1}_x M|\psi'\rangle, \tag{7}$$

where $|\psi'\rangle$ is the state obtained after the superposition state passing through the quantum circuit. To analyze the minimum value of Eq. (7) and understand the interference between data, let us first consider the quantum state of each $|x^i\rangle$ after it passes through the Rot gate, which can be written as $|x'^i\rangle = \alpha'_i|0\rangle + \beta'_i|1\rangle$ in general. Consequently, the superposition state after the Rot gate is expressed as $[(\alpha'_{14}|0\rangle + \beta'_{14}|1\rangle)|00\rangle + (\alpha'_{23}|0\rangle + \beta'_{23}|1\rangle)|01\rangle]/N$, where $\alpha'_{ij} = \alpha'_i + \alpha'_j$, $\beta'_{ij} = \beta'_i + \beta'_j$. After this transformed superposition state passes through subsequent quantum circuits, the reduced density matrix for the final output qubit and label qubit is given by

$$\begin{aligned}\widetilde{\rho} = \Bigg[&\left(|\alpha'_{14}|^2\cos^2\frac{b}{2} + |\beta'_{14}|^2\cos^2\frac{w_1+b}{2}\right)|00\rangle\langle 00|\dots\\ &+\left(|\alpha'_{23}|^2\cos^2\frac{b}{2} + |\beta'_{23}|^2\cos^2\frac{w_1+b}{2}\right)|01\rangle\langle 01|\dots\\ &+\left(|\alpha'_{14}|^2\sin^2\frac{b}{2} + |\beta'_{14}|^2\sin^2\frac{w_1+b}{2}\right)|10\rangle\langle 10|\dots\\ &+\left(|\alpha'_{23}|^2\sin^2\frac{b}{2} + |\beta'_{23}|^2\sin^2\frac{w_1+b}{2}\right)|11\rangle\langle 11|\cdots\Bigg]\Big/N^2,\end{aligned} \tag{8}$$

and the loss function of Eq. (7) can then be expressed as $L_B^{\text{Train}} = \mathrm{Tr}(\widetilde{\rho}M)$.

The above equation clearly indicates that interference between data points, as represented by the coefficients $|\alpha'_{ij}|^2$ and $|\beta'_{ij}|^2$, contributes significantly to the loss function and hence the training result. The minimum value of the loss function, specifically -1, is reached when $\alpha'_{23} = 0$, $\beta'_{14} = 0$, $b = 0$, and $w_1 = \pi$, or when $\alpha'_{14} = \beta'_{23} = 0$, $b = w_1 = \pi$. In the first solution, where $\alpha'_{23} = \beta'_{14} = 0$, we can further derive the cosines of the polar angles of both $|x'^1\rangle$ and $|x'^4\rangle$ ($|x'^2\rangle$ and $|x'^3\rangle$) to be $1/\sqrt{3}$ ($-1/\sqrt{3}$) (see Appendix B), and the second solution is analogous. This immediately prove that the solutions corresponding to the minima in both batch input and individual input are identical. In other words, it implies that training with a superposition input can at least achieve the same accuracy in the current classification task as with individual inputs.

To validate the above analytic results, we simulate the quantum circuit depicted in Fig. 3(b) by utilizing PennyLane [26]. In the simulation, we set $\{|x^i\rangle\}_{i=1}^4 = \{|0\rangle, \sqrt{1/3}|0\rangle + \sqrt{2/3}|1\rangle, \sqrt{1/3}i|0\rangle + \sqrt{2/3}ie^{i\frac{4\pi}{3}}|1\rangle, \sqrt{1/3}|0\rangle + \sqrt{2/3}e^{i\frac{2\pi}{3}}|1\rangle\}$ [see also Fig. 3(a)]. Note that this fixes $\gamma_{14} = \gamma_{23} = 0$. We employ Adam as the optimizer, using the same initialization parameters for both individual and batch inputs. The simulation results are presented in Fig. 3(c). It can be observed that the minimum values to which L_I^{NO} and L_B^{Train} converge are consistent with the theoretical analysis. In the superposition input case, we further define the test loss function precisely aligned with Eq. (5),

$$L_B^{\text{Test}} = \frac{1}{4}\sum_i^4 \langle\hat{y}^i y^i|M|\hat{y}^i y^i\rangle, \tag{9}$$

with individual data points $\{|x^i\rangle|y^i\rangle\}_{i=1}^4$. We observe a faster convergence to the expected minimum in L_B^{Test} compared with L_I^{NO}, indicating an equally accurate but more efficient training with the superposition input. We note, however, that the performance of the batch input can be influenced by the relative phases γ_{14} and γ_{23}. We will address this issue systematically in our future work.

To visualize the decision boundary formed by the trained quantum circuit, we sample over the Bloch sphere and use the single-qubit states as inputs to the trained circuit. Upon measuring the output qubit, the expectation value falls within $[-1, 1]$. This measurement result is then used to color each corresponding point on the Bloch sphere, as shown in Fig. 3(d). Meanwhile, we plot the regular tetrahedron for the data points after passing through the Rot gate. The visualization confirms that the circuit effectively addresses the XOR problem. In classical approaches, data from the XOR problem are often visualized on a two-dimensional (2D) plane, where the four distinct points form a grid. This grid representation can be viewed as a projection of our tetrahedron representation onto a plane. Although both underlying spaces, the plane and the Bloch sphere, are two-dimensional, the latter clearly offers better separability in this problem.

We close this section with one general remark that is well illustrated by studying the exactly solvable XOR problem: compared with the orthogonal encoding scheme, nonorthogonal encoding can be loosely regarded as data compression, which apparently will lead to inevitable weakening of the training performance, as, for example, the significantly lifted minimum value of the loss function L_I^{NO} versus L_I^{O} suggests; quantum superposition, however, when exploited in the input state, can introduce interaction among data by way of quantum interference [cf. Eq. (8)], hence admitting potential advantages that override the apparent weakness to result in more efficient training as seen in our example here.

III. DATA PARALLELISM WITH GENERIC QNNs

In line with the broad definition of QNNs established earlier, we introduce a generic quantum circuit framework as depicted in Fig. 4, assuming that data have already been encoded into quantum states and focusing on how to utilize these quantum states. Based on this layout, we discuss quantum data parallelism. A wide variety of QNNs [17,18,25,27–31], including those implemented with repeat-until-success (RUS) circuits while preserving unitarity [18,25,27], can be accommodated within this framework. Moreover, direct measurement of input data qubits [28–30] can always be recast into measurement of extra output qubits controlled pairwise

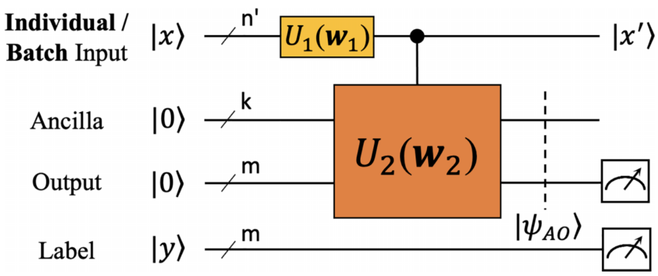

FIG. 4. A general quantum circuit framework that accommodates a variety of QNNs. Quantum data parallelism is explored by considering a superposition initial state $\sum_i \sqrt{v^i}e^{i\gamma_i}|x^i\rangle|y^i\rangle$, where each $|x^i\rangle|y^i\rangle$ encodes a pair of data x^i and label y^i.

by the input data qubits with CNOT gates, thus making our framework more inclusive.

Consider a classical supervised dataset $\{(x^i, y^i)\}_{i\in T}$ with paired data $x^i \in \mathbb{C}^n$ and label $y^i \in \{0, 1\}^m$. We compare two training approaches: individual input and batch input. In the individual input approach, each training data point from the classical dataset can be encoded into a quantum state $|x^i\rangle|y^i\rangle$. In the batch input approach, a finite subset B, which is possibly the full set T, of the training dataset can be encoded as the superposition state $\sum_{i\in B}\sqrt{v^i}e^{i\gamma_i}|x^i\rangle|y^i\rangle$, where the weight $\sqrt{v^i}$ (with $\sum_{i\in B} v^i = 1$ and $v^i \geqslant 0$) and the relative phase γ_i can be freely chosen. In the preceding section on the XOR problem, we assumed equal weights for each data point and set the relative phase to $e^{i\gamma_i} = 1$ for simplicity. By including variables v^i and γ_i, an expanded and flexible representation space can be explored.

As illustrated in Fig. 4, the encoded input states are transformed unitarily, with a parameter set $\boldsymbol{w}_1$, before acting as control to the unitary transformation of the ancilla and the output qubits, with another parameter set $\boldsymbol{w}_2$. Let $|x'^i\rangle|\psi^i_{AO}\rangle|y^i\rangle$ denote the final state for the initial input state $|x^i\rangle|0\rangle^{\otimes(k+m)}|y^i\rangle$, where A and O represent the Hilbert space of ancilla and output qubits, respectively. For the training process, the agreement between the output qubits and the label qubits is evaluated by averaged pairwise ZZ measurements

$$M = -\frac{1}{m}\sum_j^m Z_{o_j}Z_{y_j}, \tag{10}$$

where o_j and y_j stand for the jth output and label qubits, respectively. The measured value of M clearly falls between -1 and 1, with -1 signifying perfect agreement. This framework, supported by the use of a label qubit, establishes a unified approach applicable to both orthogonal and nonorthogonal cases, naturally accommodating both binary and multiclass scenarios.

In the individual input approach, for a pair of data and the label $|x^i\rangle|y^i\rangle$, the reduced density matrix of subsystem AY is

$$\rho^i_{AY} = |\psi^i_{AO}y^i\rangle\langle\psi^i_{AO}y^i|, \tag{11}$$

where Y represents the space of output qubits and label qubits. Note that the reduction of the subsystem for $|x'\rangle$ is trivial here because they have served as control qubits only and the direct product structure is preserved. For the full dataset, we define the following loss function: $L_I = \sum_{i\in T} v^i \mathrm{Tr}(\rho^i_Y M)$, where $\rho^i_Y = \mathrm{Tr}_A\rho^i_{AY}$, with Tr_A the partial trace over the subspace for ancilla qubits.

In the batch input approach, the superposition input state is expressed as $\sum_{i\in B}\sqrt{v^i}e^{i\gamma_i}|x^i\rangle|y^i\rangle$. As we will discuss later, the entanglement here between data qubits and label qubits plays a crucial role in quantum parallelism. For this state, the full density matrix for the final state of the QNN is given by

$$\begin{aligned}\rho^S &= \left(\sum_i \sqrt{v^i}e^{i\gamma_i}|x'^i\psi^i_{AO}y^i\rangle\right)\\ &\quad\times\left(\sum_j \sqrt{v^j}e^{-i\gamma_j}\langle x'^j\psi^j_{AO}y^j|\right)\\ &= \sum_{i,j}\sqrt{v^iv^j}e^{i(\gamma_i-\gamma_j)}|x'^i\psi^i_{AO}y^i\rangle\langle x'^j\psi^j_{AO}y^j|,\end{aligned} \tag{12}$$

which implies that the reduced density matrix for subsystem AY is

$$\begin{aligned}\rho^S_{AY} &= \mathrm{Tr}_X(\rho^S)\\ &= \sum_{i,j}\sqrt{v^iv^j}e^{i(\gamma_i-\gamma_j)}\langle x'^j|x'^i\rangle|\psi^i_{AO}y^i\rangle\langle\psi^j_{AO}y^j|.\end{aligned} \tag{13}$$

Note that, unlike in the individual input approach, here the subsystem for $|x'\rangle$ and subsystem AY are highly entangled. For the batch input, we define the loss function to be $L_B = \mathrm{Tr}(\rho_Y M)$, where $\rho_Y = \mathrm{Tr}_A\rho^S_{AY}$.

In general, estimating the loss function L_I and L_B with error ϵ requires approximately $O(\frac{1}{\epsilon^2})$ measurements. Further optimization of the loss functions can be achieved through both gradient-based and gradient-free algorithms. Automatic differentiation schemes [32] can be used to calculate gradients in simulations. In practice, it is found that the simultaneous perturbation stochastic approximation (SPSA) performs well in small-scale noisy experimental settings [28,29]. For testing, we directly measure the output qubits and use the outcome as the predicted labels. The efficacy of these predictions is quantified by the accuracy across the test set $\{(x^j, y^j)\}_{j=1}^{T_{\mathrm{test}}}$, defined as $\mathrm{acc} = \frac{1}{T_{\mathrm{test}}}\sum_j I[\mathrm{Tr}(\rho^j_Y M) \leqslant 0]$, where $I(\cdot)$ equals 1 when the jth data are correctly classified and 0 otherwise.

Having established the generic quantum circuit framework for QNNs, we now turn our attention to quantum data parallelism. A straightforward advantage of employing a superposition state as the input in our framework is that the state of the output qubits can be computed in parallel for all the training data points encoded in the superposition. We emphasize, however, that this apparent advantage is only fulfilled when the measurement of the final states satisfies two criteria: first, it efficiently yields an average over the superposed training data; second, it effectively implements a workable loss function. The first criterion is obviously satisfied by the measurement of M in Eq. (10) as it directly measures each combination of $\{Z_{o_j}\}$ and $\{Z_{y_j}\}$ eigenvalues with probability given by the sum over corresponding data points. It is worth noting that the minimum $\langle M\rangle$ for the batch input implies the maximum possible entanglement between the output and the label qubits. In this sense, to find parameters $\boldsymbol{w}_1$ and $\boldsymbol{w}_2$ that optimize the loss function L_B can be regarded as finding a map from $x^i \in \mathbb{C}^n$ to $o^i \in \{0, 1\}^m$ that maximizes (up to a basis transformation) the entanglement between the output qubits $\{o_j\}$ and the label qubits $\{y_j\}$. For the second criterion, we demonstrate its validity by proving the following theorem that connects the loss functions for the individual input and the batch input with orthogonal encoding, as well as by explicit numerical simulations of real-world problems with both orthogonal and nonorthogonal encoding.

Theorem 1 (Data parallelism by quantum superposition). If the input data are encoded into a set of orthonormal quantum states $\{|x^i\rangle\}_{i=1}^T$ (i.e., $\langle x^i|x^j\rangle = \delta_{ij}$), and the loss function for each individual pair of data and the label (x^i, y^i) can be obtained in the above-defined quantum neural network by a joint measurement on the output qubits and the label qubits through a linear operator M [i.e., $l^i := \mathrm{Tr}(\rho_Y^i M)$, with ρ_Y the reduced density matrix for the subsystem Y composed of the output qubits and the label qubits], then the overall loss function defined as a weighted average $L_I := \sum_i^T (v^i l^i)$ with $\sum_i^T v^i = 1$ and $v^i \geqslant 0$ can be obtained exactly by the joint measurement of M with respect to a single superposition input state $\sum_{i=1}^T \sqrt{v^i} e^{i\gamma_i} |x^i\rangle|y^i\rangle$, such that $L_B := \mathrm{Tr}(\rho_Y M) = L_I$.

Proof. According to Eq. (13) and utilizing the orthonormal condition $\langle x'^i|x'^j\rangle = \langle x^i|x^j\rangle = \delta_{ij}$, we obtain

$$\begin{aligned}\rho_{AY}^S &= \sum_{i,j} \sqrt{v^i v^j} e^{i(\gamma_i - \gamma_j)} \langle x'^j|x'^i\rangle \big|\psi_{AO}^i y^i\big\rangle\big\langle\psi_{AO}^j y^j\big| \\ &= \sum_i v^i \big|\psi_{AO}^i y^i\big\rangle\big\langle\psi_{AO}^i y^i\big| = \sum_i v^i \rho_{AY}^i. \end{aligned} \tag{14}$$

It immediately follows that

$$\begin{aligned} L_B :=& \mathrm{Tr}(\rho_Y M) = \mathrm{Tr}\big[\big(\mathrm{Tr}_A \rho_{AY}^S\big) M\big] \\ =& \sum_i^T v^i \mathrm{Tr}\big[\big(\mathrm{Tr}_A \rho_{AY}^i\big) M\big] = \sum_i^T v^i \mathrm{Tr}\big(\rho_Y^i M\big) = L_I. \end{aligned} \tag{15}$$

■

With nonorthogonal encoding, by contrast, Eq. (14) is invalidated, leading to a situation in which L_B is no longer equivalent to L_I. This can be seen as the interference between distinct data points, represented by nonvanishing $\langle x'^j|x'^i\rangle$ for $j \neq i$, becomes manifested in the loss function for batch input. Particularly, if the measurement M is diagonal with respect to the labels $|y^i\rangle$, as, for example, in Eq. (10), the interference effectively occurs only between data points associated with the same label. Such an interference effect, while making the interpretation of the loss function L_B less transparent in general, may hold a potential advantage over the orthogonal encoding scheme when properly exploited. This remains an intriguing direction for our future investigations.

IV. NUMERICAL SIMULATIONS

To assess the proposed QNN framework and its capability to achieve data parallelism, we conduct numerical simulations on three distinct problems: the parity problem, the Iris dataset, and the MNIST dataset. The parity problem, which focuses on learning a Boolean function from binary inputs, is particularly suitable for basis encoding and thus serves as a test problem for examining the QNN's performance in the orthogonal case. The Iris dataset, commonly used for categorizing Iris flower species, is employed to assess the QNN's parallelism in the nonorthogonal case. Finally, the MNIST dataset, a standard benchmark for handwritten digit recognition, provides a comprehensive platform for comparing the QNN's performance across four different settings: orthogonal versus nonorthogonal encoding and individual versus batch input.

As benchmarks, we train classical feedforward neural networks (FNNs) on each of the problems. Our simulations of both QNNs and FNNs are carried out using the Pennylane [26] and TensorFlow [33,34] programming frameworks. We consistently use Adam as our optimizer for all experiments. Our code is available at [35].

A. Parity

Parity is a function $f(x)$ that maps an n-bit binary vector $x \in \{0, 1\}^n$ to a binary output y. The output y is 1 if the number of 1's in x is odd, and 0 if it is even. In our experiment, we use $n = 10$, resulting in a total of 1024 data points, of which 80% are used for training and 20% for testing. The QNN we use is similar to the one shown in Fig. 5(a), without the U_1 part and with two ancilla qubits in the U_2 part. The FNN used for comparison consists of three layers: the first two are fully connected layers with 500 and 100 neurons, respectively, following a structure similar to those in [36,37]. At this point, the number of parameters in the FNN is more than 1900 times that of the QNN. During training, the learning rate for the QNN is set to 0.05 and for the FNN to 0.01, while the batch size for both models is set to 64. The results are shown in Fig. 5(c).

B. Iris

Iris consists of 150 samples from three species of Iris flowers, each described by four continuous features: sepal length, sepal width, petal length, and petal width, all measured in centimeters. Here, we train the model to recognize the first two classes of flowers in the Iris dataset, utilizing an even split of data from each class for both training and testing. Due to the relative simplicity of this problem and the limited amount of training data, the QNN employed in our study omits the use of U_1 and does not incorporate ancilla qubits in U_2. The FNN, designed for comparison with the QNN, is structured with a similar number of parameters. It includes a flattened layer that reshapes the input data into a one-dimensional array, followed by a single dense layer with one neuron. During training, the learning rates for both the QNN and FNN are set to 0.05. Given the dataset's size, we opt for batch training using the entire training data. We conduct five trials for each model with random initializations, maintaining identical initialization parameters for both nonparallel and parallel QNN training approaches in each trial. The results of these experiments, depicted in Fig. 5(d), demonstrate the comparative performance of both models.

C. MNIST

We adopt a downsampled version of the MNIST, which consist of 4×4 images. The QNN used for classification is depicted in Fig. 5(a). In the orthogonal case, we binarize each

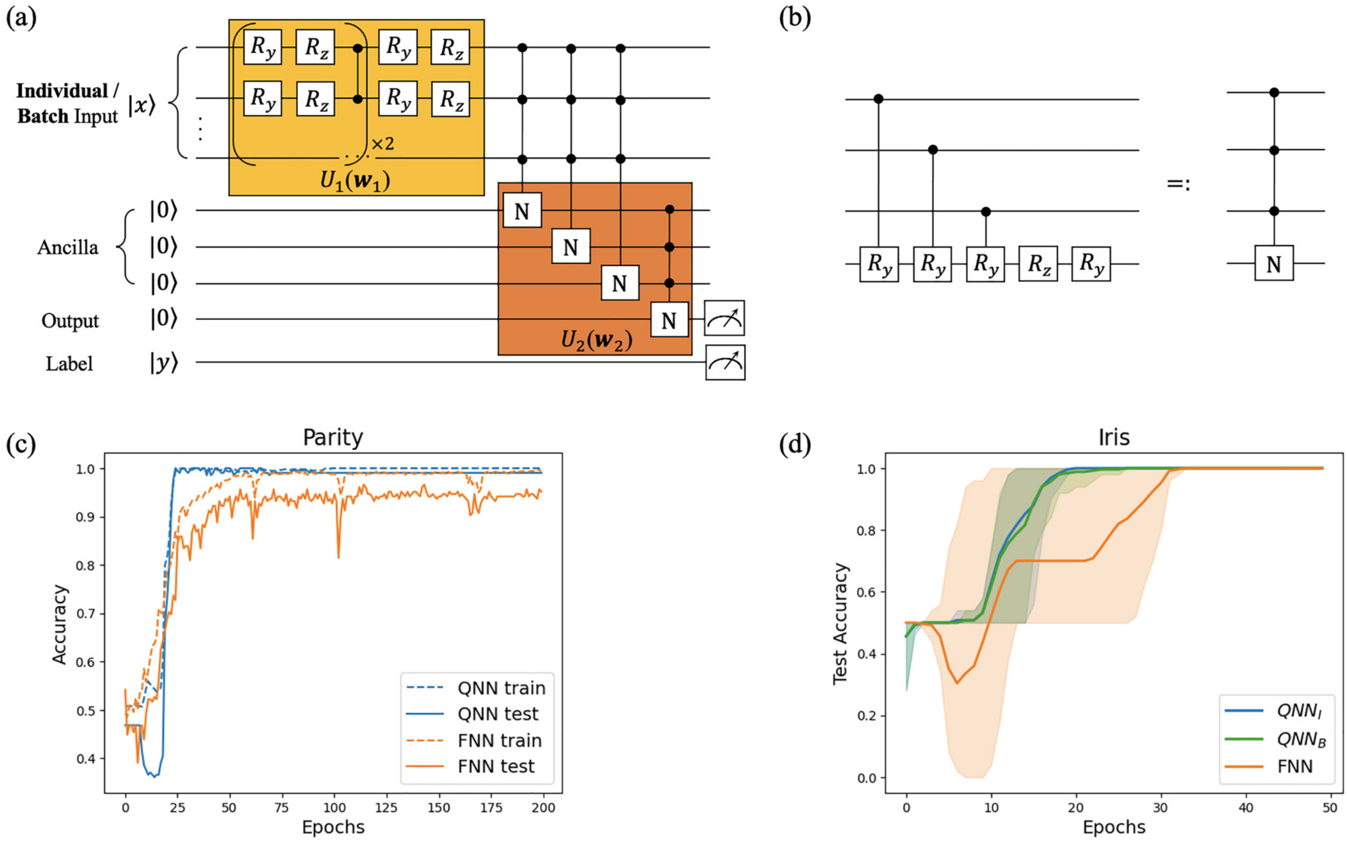

FIG. 5. Numerical simulations to verify and evaluate our quantum parallelization schemes. (a) General quantum circuit architecture employed in the simulations. (b) Decomposition of the N Gate, inspired by one construction of quantum neurons [18,24,25]. Our design omits the RUS circuit, as a steep nonlinear activation function does not serve our purpose. (c) Evolution of model accuracy in solving parity problems, with blue and orange lines delineating quantum (QNN) and classical (FNN) models, respectively. (d) Averaged test accuracy progression in Iris classification, where the average is plotted as solid lines and the shaded area denotes the range of test accuracies observed across five separate experiments. The mostly overlapping blue and green lines correspond to nonparallel and parallel trainings of the QNN, respectively.

image and use basis encoding to convert it into a quantum state; U_1 is not employed. According to Theorem 1, individual and batch training are equivalent. Hence, we present only individual training simulations. For the nonorthogonal case, binarization is not required and amplitude encoding is adopted. Here, quantum interference in parallel processing necessitates separate training for individual and batch inputs. In our setup, we choose equal weights for each data point and set all relative phases to zero. As a benchmark, FNNs are employed, adopting a structure similar to the U_2 in Fig. 5(a) with four neurons and two layers, both for binarized and nonbinarized images. For both the QNN and the FNN, the learning rates are set to 0.01, with each model employing a batch size of 128. The comparison with FNNs across five experiments is listed in Table I. The primary aim of the simulation is to verify the feasibility of parallel computation in QNNs, rather than to demonstrate a quantum advantage. Consequently, we do not extensively explore various *Ansätzes* to showcase accuracy advantages on the MNIST dataset.

V. DISCUSSION

Demonstrated by both analytical and numerical means, we have established that quantum-superposition-based data parallelism in QNNs is not only feasible but also remarkably effective, irrespective of the orthogonality of input quantum states. Here, we remark on several fundamental and interconnected open questions.

In this work, our primary goal is to investigate how quantum superposition facilitates data parallelism in QNNs. As such, we did not delve into the problem of preparing the input superposition states. Creating and maintaining specific superposition states necessarily incurs a cost: the exact preparation of a generic n-qubit state requires exponentially circuit depth in the worst case [38–41]. Recently, several near-term techniques have been proposed, including

TABLE I. Classification results for digits 3 and 6 on MNIST dataset.

Model	Input	Qubits	Params	Test Accuracy
QNN	Orthogonal	20	59	91.5%
FNN	N/A	N/A	55	92.8%
QNN	Nonorthogonal (Individual)	9	47	97.8%
QNN	Nonorthogonal (Batch)	9	47	97.8%
FNN	N/A	N/A	55	98.3%

methods based on numerical integration [42,43] and quantum generative adversarial networks (QGANs) [44,45], which construct $O[poly(n)]$-sized quantum circuits that approximate learned distributions. Additionally, fault-tolerant approaches to quantum state preparation have been developed, offering theoretical guarantees in terms of convergence, circuit depth, and the number of ancillary qubits required [46–48]. For example, Zhang *et al.* [47] recently demonstrated that any n-qubit quantum state can be prepared with a circuit depth of $\Theta(n)$, though this requires an exponential number of ancillary qubits. Furthermore, many practically relevant quantum states can be prepared more efficiently by leveraging their special properties or sparsity, further reducing the necessary resources [47,49,50]. Although preparing arbitrary superposition states is challenging, our parallel approach can still gain advantages for certain tasks where the computational complexity involved in processing data surpasses that for generating the required superposition states. In this regard, efficient algorithms to encode a classical dataset into a quantum superposition state become a crucial associated direction to be further explored.

A closely related issue can be phrased as the compression problem, that is, to determine the optimal number of qubits necessary to represent a dataset in a specific type of tasks. For a given classical dataset, orthogonal encoding presumably requires a maximum number of qubits while maintaining the full information in the sense that any superposition state can be uniquely decomposed to its original data; nonorthogonal encoding, by comparison, requires fewer qubits and apparently suffers from a loss of full separability. This loss can generally be viewed as the loss of information due to data compression, which increases if the number of qubits decreases further. When a specific type of task is targeted, however, the lost information in certain compression schemes can be insignificant in terms of the outcome accuracy, as the examples in this work have already demonstrated. From a distance perspective, the distance between any two data points remains constant in orthogonal encoding, while nonorthogonal encoding—like word2vec embeddings in classical NLP—offers a more compact representation where similar data points are encoded with smaller relative distances. More importantly, by exploiting the extra degrees of freedom in the relative phases associated with different data points and the quantum interference effect thereof, both of which are missing with orthogonal encoding, compressed encoding may even offer critical quantum advantages in solving certain learning problems. This poses another theoretical challenge to derive a lower bound of the qubit number and an optimal scheme of compression for a given accuracy goal based on quantum information theory.

In training the QNN models, we have mentioned that to optimize the loss function with superposition input is to maximize the entanglement between the output and the label qubits. This again can be recast into an information theoretical problem about the quantum circuit complexity for implementing an entanglement transfer from the input data qubits, which are maximally (up to predetermined data composition) entangled with the label qubits by construction, to the output qubits. Solving such a problem will certainly shed light on our understanding and application of QNNs.

In summary, our work serves as a pioneering, proof-of-principle study of data parallelism in QNNs. The realization of this quantum parallelism can clearly be attributed to the fundamental principles of quantum superposition and quantum entanglement, bolstered by the capability of quantum measurement to extract global information effectively. This study paves the way for a new paradigm for parallel data processing in quantum machine learning.

APPENDIX A: INDIVIDUAL INPUT FOR XOR IN THE NONORTHOGONAL ENCODING

In this Appendix, we theoretically derive the minimum value of Eq. (6). Since w_1 and b are independent of the other parameters, both $\cos\theta_2+\cos\theta_3-\cos\theta_1-\cos\theta_4$ and $\cos b-\cos(w_1+b)$ can achieve their extreme value independently. Initially, we assume $\cos b-\cos(w_1+b)=2$, resulting in $b=0$ and $w_1=\pi$. Subsequently, we explore the minimum value of $\cos\theta_2+\cos\theta_3-\cos\theta_1-\cos\theta_4$. Another scenario is when $\cos b-\cos(w_1+b)=-2$. In this case, we would need to explore the maximum value of $\cos\theta_2+\cos\theta_3-\cos\theta_1-\cos\theta_4$.

To determine the extremum of $\cos\theta_2+\cos\theta_3-\cos\theta_1-\cos\theta_4$, we transform the regular tetrahedron in the two-dimensional complex plane into Cartesian coordinates. The following vectors define the four vertices of a tetrahedron with edge length 2, centered at the origin, and two level edges:

$$\vec{t}_1=\begin{bmatrix}\sqrt{2}\\0\\1\end{bmatrix},\quad \vec{t}_2=\begin{bmatrix}0\\\sqrt{2}\\-1\end{bmatrix},\quad \vec{t}_3=\begin{bmatrix}0\\-\sqrt{2}\\-1\end{bmatrix},\quad \vec{t}_4=\begin{bmatrix}-\sqrt{2}\\0\\1\end{bmatrix}.$$

Let us consider an arbitrary rotation applied to the tetrahedron. The problem of finding the extremum thus translates to determining which rotation matrix allows the expression to attain its extremum. Any 3D rotation can be decomposed into a ZXZ Euler angle sequence, using three rotations: first around the z-axis, then the rotated x-axis, and finally the twice-rotated z-axis. Since $\cos\theta_2+\cos\theta_3-\cos\theta_1-\cos\theta_4$ is only related to θ, the final rotation along the z-axis does not affect the extremum and can be disregarded. Thus, the rotation matrix can be represented as

$$\begin{aligned} r_x(\phi_1)r_z(\phi_2) &= \begin{bmatrix}1&0&0\\0&\cos\phi_1&-\sin\phi_1\\0&\sin\phi_1&\cos\phi_1\end{bmatrix}\begin{bmatrix}\cos\phi_2&-\sin\phi_2&0\\\sin\phi_2&\cos\phi_2&0\\0&0&1\end{bmatrix}\\ &= \begin{bmatrix}\cos\phi_2&-\sin\phi_2&0\\\cos\phi_1\sin\phi_2&\cos\phi_1\cos\phi_2&-\sin\phi_1\\\sin\phi_1\sin\phi_2&\sin\phi_1\cos\phi_2&\cos\phi_1\end{bmatrix}.\end{aligned}$$

Applying the rotation matrix to the four vertices, we obtain

$$\vec{t}_1'=\begin{bmatrix}\sqrt{2}\cos\phi_2\\\sqrt{2}\cos\phi_1\sin\phi_2-\sin\phi_1\\\sqrt{2}\sin\phi_1\sin\phi_2+\cos\phi_1\end{bmatrix},$$

$$\vec{t}_2'=\begin{bmatrix}-\sqrt{2}\sin\phi_2\\\sqrt{2}\cos\phi_1\cos\phi_2+\sin\phi_1\\\sqrt{2}\sin\phi_1\cos\phi_2-\cos\phi_1\end{bmatrix},$$

$$\vec{t}_3' = \begin{bmatrix} \sqrt{2}\sin\phi_2 \\ -\sqrt{2}\cos\phi_1\cos\phi_2 + \sin\phi_1 \\ -\sqrt{2}\sin\phi_1\cos\phi_2 - \cos\phi_1 \end{bmatrix},$$

$$\vec{t}_4' = \begin{bmatrix} -\sqrt{2}\cos\phi_2 \\ -\sqrt{2}\cos\phi_1\sin\phi_2 - \sin\phi_1 \\ -\sqrt{2}\sin\phi_1\sin\phi_2 + \cos\phi_1 \end{bmatrix}.$$

Given that $\cos\theta = \frac{\vec{z}\cdot\vec{t}}{\sqrt{3}}$, where $\vec{z} = [0, 0, 1]$, we can deduce

$$\cos\theta_1 = \frac{\sqrt{2}\sin\phi_1\sin\phi_2 + \cos\phi_1}{\sqrt{3}},$$

$$\cos\theta_2 = \frac{\sqrt{2}\sin\phi_1\cos\phi_2 - \cos\phi_1}{\sqrt{3}},$$

$$\cos\theta_3 = \frac{-\sqrt{2}\sin\phi_1\cos\phi_2 - \cos\phi_1}{\sqrt{3}},$$

$$\cos\theta_4 = \frac{-\sqrt{2}\sin\phi_1\sin\phi_2 + \cos\phi_1}{\sqrt{3}}.$$

Utilizing this, we derive

$$\cos\theta_2 + \cos\theta_3 - \cos\theta_1 - \cos\theta_4 = -\frac{4\cos\phi_1}{\sqrt{3}}.$$

This expression attains its minimum value $-\frac{4}{\sqrt{3}}$ when $\phi_1 = 0$ and its maximum value $\frac{4}{\sqrt{3}}$ when $\phi_1 = \pi$, and it is independent of ϕ_2. That is, when $\cos\theta_2 = \cos\theta_3 = -\cos\theta_1 = -\cos\theta_4 = -\frac{1}{\sqrt{3}}$, $\cos\theta_2 + \cos\theta_3 - \cos\theta_1 - \cos\theta_4$ attains its minimum value of $-\frac{4}{\sqrt{3}}$. For all values equal to $\frac{1}{\sqrt{3}}$, it reaches its maximum value of $\frac{4}{\sqrt{3}}$. This implies that when the edges connecting two data points with the same label are parallel to the *xoy* plane, Eq. (6) can attain its minimum value $-\frac{1}{\sqrt{3}}$.

APPENDIX B: BATCH INPUT FOR XOR IN THE NONORTHOGONAL ENCODING

In this Appendix, we theoretically derive the minimum value of Eq. (7). The transformed superposition state $[(\alpha_{14}'|0\rangle + \beta_{14}'|1\rangle)|00\rangle + (\alpha_{23}'|0\rangle + \beta_{23}'|1\rangle)|01\rangle]/N$ evolves into

$$\begin{aligned}\Bigg[&\left(\alpha_{14}'\cos\frac{b}{2}|00\rangle + \alpha_{14}'\sin\frac{b}{2}|01\rangle\right)|0\rangle \\ &+ \left(\beta_{14}'\cos\frac{w_1+b}{2}|10\rangle + \beta_{14}'\sin\frac{w_1+b}{2}|11\rangle\right)|0\rangle \\ &+ \left(\alpha_{23}'\cos\frac{b}{2}|00\rangle + \alpha_{23}'\sin\frac{b}{2}|01\rangle\right)|1\rangle \\ &+ \left(\beta_{23}'\cos\frac{w_1+b}{2}|10\rangle + \beta_{23}'\sin\frac{w_1+b}{2}|11\rangle\right)|1\rangle\Bigg]\Big/N\end{aligned}$$

after passing through controlled-$R_y(w_1)$ and $R_y(b)$. The density matrix of this state is

$$\begin{aligned}\rho = \Bigg[\Bigg(&|\alpha_{14}'|^2\cos^2\frac{b}{2}|000\rangle\langle 000| \\ &+ |\alpha_{14}'|^2\cos\frac{b}{2}\sin\frac{b}{2}|000\rangle\langle 010| + \cdots \\ &+ |\beta_{14}'|^2\cos^2\frac{w_1+b}{2}|100\rangle\langle 100| + \cdots\Bigg)\Bigg]\Big/N^2\cdots.\end{aligned}$$

The reduced density matrix is

$$\begin{aligned}\widetilde{\rho} &= \mathrm{tr}_X(\rho) \\ &= \Bigg[\left(|\alpha_{14}'|^2\cos^2\frac{b}{2} + |\beta_{14}'|^2\cos^2\frac{w_1+b}{2}\right)|00\rangle\langle 00|\ldots \\ &\quad + \left(|\alpha_{23}'|^2\cos^2\frac{b}{2} + |\beta_{23}'|^2\cos^2\frac{w_1+b}{2}\right)|01\rangle\langle 01|\ldots \\ &\quad + \left(|\alpha_{14}'|^2\sin^2\frac{b}{2} + |\beta_{14}'|^2\sin^2\frac{w_1+b}{2}\right)|10\rangle\langle 10|\ldots \\ &\quad + \left(|\alpha_{23}'|^2\sin^2\frac{b}{2} + |\beta_{23}'|^2\sin^2\frac{w_1+b}{2}\right)|11\rangle\langle 11|\cdots\Bigg]\Big/N^2.\end{aligned}$$

We denote the diagonal element of the matrix $\widetilde{\rho}$ at row i and column i as $\widetilde{\rho}_i$. For example, $\widetilde{\rho}_{00} = [|\alpha_{14}'|^2\cos^2\frac{b}{2} + |\beta_{14}'|^2\cos^2\frac{w_1+b}{2}]/N^2$. Let $\varrho_0 = \widetilde{\rho}_{00} + \widetilde{\rho}_{11}$ and $\varrho_1 = \widetilde{\rho}_{01} + \widetilde{\rho}_{10}$. We have

$$\begin{aligned}&\varrho_0 + \varrho_1 = 1, \\ &1 \geqslant \varrho_0, \quad \varrho_1 \geqslant 0.\end{aligned}$$

The loss function is

$$L_B^{\text{Train}} = \mathrm{tr}(\widetilde{\rho}M) = \varrho_1 - \varrho_0 \geqslant -\varrho_0 \geqslant -1.$$

The inequality attains equality when $\varrho_1 = 0$ and 1. $\varrho_1 = 0$ implies $\widetilde{\rho}_{01} = \widetilde{\rho}_{10} = 0$, leading to two solutions,

$$\begin{aligned}&\alpha_{23}' = 0, \quad \cos\frac{w_1+b}{2} = 0, \\ &\beta_{14}' = 0, \quad \sin\frac{b}{2} = 0; \\ &\beta_{23}' = 0, \quad \cos\frac{b}{2} = 0, \\ &\alpha_{14}' = 0, \quad \sin\frac{w_1+b}{2} = 0.\end{aligned} \tag{B1}$$

The analytical approaches for both solutions are similar. In this Appendix, we will focus on the first case. Based on the left solution, the four states can be expressed as

$$|x'^1\rangle = e^{i\gamma_1}\left[e^{-i\varphi_1}\cos\frac{\theta_1}{2}, \sin\frac{\theta_1}{2}\right]^T,$$

$$|x'^2\rangle = e^{i\gamma_2}\left[\cos\frac{\theta_2}{2}, e^{i\varphi_2}\sin\frac{\theta_2}{2}\right]^T,$$

$$|x'^3\rangle = e^{i\gamma_2}\left[-\cos\frac{\theta_2}{2}, e^{i\varphi_3}\sin\frac{\theta_2}{2}\right]^T,$$

$$|x'^4\rangle = e^{i\gamma_1}\left[e^{-i\varphi_4}\cos\frac{\theta_1}{2}, -\sin\frac{\theta_1}{2}\right]^T,$$

where $1 \geqslant \cos\frac{\theta_1}{2}, \cos\frac{\theta_2}{2}, \sin\frac{\theta_1}{2}, \sin\frac{\theta_2}{2} \geqslant 0$. Based on these, we can compute their inner product

$$\begin{aligned}
\langle x'^2|x'^1\rangle &= e^{i(\gamma_1-\varphi_1-\gamma_2)}\bigg(\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2} \\
&\quad + e^{i(\varphi_1-\varphi_2)}\sin\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\bigg), \\
\langle x'^2|x'^4\rangle &= e^{i(\gamma_1-\varphi_4-\gamma_2)}\bigg(\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2} \\
&\quad + e^{i(\varphi_4-\varphi_2+\pi)}\sin\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\bigg), \\
\langle x'^3|x'^1\rangle &= e^{i(\gamma_1-\varphi_1-\gamma_2+\pi)}\bigg(\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2} \\
&\quad + e^{i(\varphi_1-\varphi_3+\pi)}\sin\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\bigg), \\
\langle x'^3|x'^4\rangle &= e^{i(\gamma_1-\varphi_4-\gamma_2+\pi)}\bigg(\cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2} \\
&\quad + e^{i(\varphi_4-\varphi_3)}\sin\frac{\theta_1}{2}\sin\frac{\theta_2}{2}\bigg).
\end{aligned}$$

Due to the nature of the regular tetrahedron, the magnitudes of these inner products should all be equal to $\frac{1}{\sqrt{3}}$. There are two possibilities for $|\langle x'^2|x'^1\rangle| = |\langle x'^2|x'^4\rangle|$, namely $\varphi_1-\varphi_2 = \varphi_4-\varphi_2+\pi$ or $\varphi_1+\varphi_4-2\varphi_2=\pi$. However, the first possibility can be ruled out, because it implies that $\varphi_1-\varphi_4=\pi$. In this case, $|x'^1\rangle$ and $|x'^4\rangle$ would be the same state (differing only by a global phase), which contradicts their representation as two distinct vertices of the regular tetrahedron. Similarly, based on $|\langle x'^3|x'^1\rangle| = |\langle x'^3|x'^4\rangle|$, we can derive $\varphi_1+\varphi_4-2\varphi_3=\pi$. From this, we infer that $\varphi_2=\varphi_3$. Furthermore, combining with $|\langle x'^2|x'^1\rangle| = |\langle x'^3|x'^4\rangle|$, we get $\varphi_1=\varphi_4$. Substituting $\varphi_1=\varphi_4$ back into $\varphi_1+\varphi_4-2\varphi_2=\pi$, we further deduce that $\varphi_1-\varphi_2=\frac{\pi}{2}$. Therefore, when $\varphi_2=\varphi_3$, $\varphi_1=\varphi_4$, and $\varphi_1-\varphi_2=\frac{\pi}{2}$, we have

$$\begin{aligned}
|\langle x'^2|x'^3\rangle| &= \left|-\cos^2\frac{\theta_2}{2}+\sin^2\frac{\theta_2}{2}\right| \\
&= |-\cos\theta_2| = \frac{1}{\sqrt{3}}, \\
|\langle x'^1|x'^4\rangle| &= \left|\cos^2\frac{\theta_1}{2}-\sin^2\frac{\theta_3}{2}\right| = |\cos\theta_1| = \frac{1}{\sqrt{3}}.
\end{aligned}$$

At this point, one should choose $\cos\theta_1=\frac{1}{\sqrt{3}}$ and $\cos\theta_2=-\frac{1}{\sqrt{3}}$, which means $\gamma_{14}=\gamma_{23}=0$. Another solution, $\cos\theta_1=-\frac{1}{\sqrt{3}}$ and $\cos\theta_2=\frac{1}{\sqrt{3}}$, is a spurious solution. Given $\alpha'_{23}=0$ and $\beta'_{14}=0$ in Eq. (B1), this second solution would cause the dominant terms to cancel out during superposition, resulting in an unreasonable superposition. Thus, the first solution for the minimal value of Eq. (7) is $\cos\theta_1=\frac{1}{\sqrt{3}}$, $\cos\theta_2=-\frac{1}{\sqrt{3}}$, $b=0$, and $w_1=\pi$. Using the same approach to analyze the second case in Eq. (B1), we can obtain the second solution $\cos\theta_1=-\frac{1}{\sqrt{3}}$, $\cos\theta_2=\frac{1}{\sqrt{3}}$, and $b=w_1=\pi$.

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